25 $a = 0,\ b = 1,\ c = 0,\ d = 1$

27 True

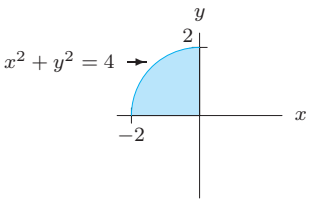
29 False

Chapter 16 Review

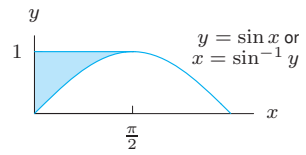
1 94.5

3 14

5



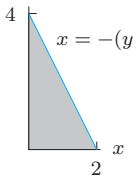
7



11 $\int_0^{2\pi}\int_0^2\int_0^3 f r\,dr dz d\theta$

13 $\int_0^{\pi/2}\int_0^{\pi}\int_0^5 f\rho^2\sin\phi\,d\rho d\phi d\theta$

15 (a) y



(b) $\int_0^2\int_0^{-2x+4} g(x,y)\,dy\,dx$

17 $10(e-2)$

19 $(1/20)\sin^5 1$

21 $(\pi/2)(1-e^{-1})$

23 1/48

25 $\int_0^{2\pi}\int_{3\pi/4}^{\pi}\int_0^{\sqrt{2}} f(\rho,\phi,\theta)\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

27 $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{-\sqrt{2-x^2-y^2}}^{-\sqrt{x^2+y^2}} h(x,y,z)\,dz\,dy\,dx$

29 Positive

31 Positive

33 Negative

35 Zero

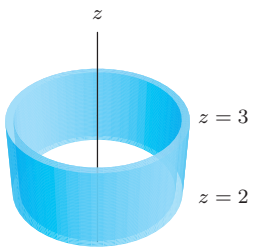
37 Positive

39 1.571

41 (a) $\int_0^{2\pi}\int_{\pi/4}^{\pi/2}\int_0^{3/\sin\theta}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

(b) 18π

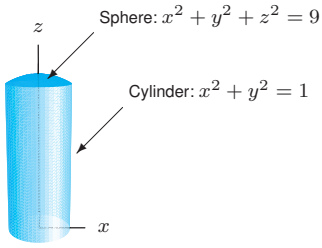
43



$\int_0^{2\pi}\int_2^3\int_{\sqrt{5}}^{\sqrt{6}} r\,dr\,dz\,d\theta$

Order of integration can be changed

45



$\int_0^{2\pi}\int_0^1\int_0^{\sqrt{9-r^2}} r\,dz\,dr\,d\theta$

47 Negative

49 Can't tell

51 Zero

53 Zero

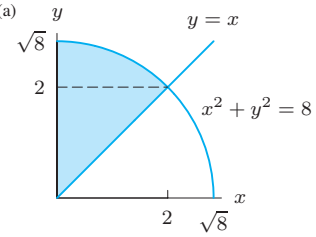
55 Zero

57 Negative

59 $\pi(3 - 2\ln 2)$

61 $162\pi/5$

63 (a)



(b) $\pi(1 - e^{-8})/8$

65 (a) $z = \sqrt{1-(y-1)^2}\quad 0 \le y \le 2\quad 0 \le x \le 10$

(b) $\int_0^{10}\int_0^2\int_0^{\sqrt{1-(y-1)^2}} f(x,y,z)\,dz\,dy\,dx$

67 (a) Half cylinder, radius 1, along x-axis, $y \ge 0, -1 \le x \le 1$

(b) $2\pi/5$

69 $\int_0^{2\pi}\int_0^4\int_3^{\sqrt{25-r^2}} r\,dz\,dr\,d\theta$
$= 52\pi/3 = 54.45\ \text{cm}^3$

71 $\int_0^{2\pi}\int_0^{\arccos(3/5)}\int_{3/\cos\phi}^5 \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$
$= 52\pi/3 = 54.45\ \text{cm}^3$

73 8π gm

75 $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{-1+\sqrt{2-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz dy dx$
$= 2\pi((7/6) - 2\sqrt{2}/3)$

77 $8\pi R^5/15$

79 (a) 3/2000

(b) 343/1000

81 $6\pi/7$

83 $a, b + c, 0$

Section 17.1

1 $x = 0, y = t, -2 \le t \le 1$

3 $x = t,\quad y = 3 - 3t,\quad 0 \le t \le 1$

5 $x = 1 + 2t,\quad y = 1 + t,\quad 0 \le t \le 1$

7 $x = t, y = 1, z = -t$

9 $x = 1,\quad y = 0,\quad z = t$

11 $x = 1 + 3t,\quad y = 2 - 3t,\quad z = 3 + t$

13 $x = -3 + 2t, y = -2 - t, z = 1 - 2t$

15 $x = 2 + 3t,\quad y = 3 - t,\quad z = -1 + t$

17 $x = 3 - 3t, y = 0, z = -5t$

19 $x = 3\cos t, y = 3\sin t, z = 5, 0 \le t < 2\pi$

21 $x = 2\cos t, y = -2\sin t, z = 0$

23 $x = 2\cos t, y = 0, z = 2\sin t$

25 $x = 0, y = 3\cos t, z = 2 + 3\sin t$

27 $x = t^2, y = t, z = 0$

29 $x = -3t^2, y = 0, z = t$

31 $x = t, y = 4 - 5t^4, z = 4$

33 $x = 3\cos t, y = 2\sin t, z = 0$

35 $x = -1 + 3t, y = 2, z = -3 + 5t$

37 $\vec{r}\,(t) = \vec{i} - 3\vec{j} + 2\vec{k} + t(3\vec{i} + 4\vec{j} - 5\vec{k}\,)$, $0 \le t \le 1, x = 1 + 3t, y = -3 + 4t, z = 2 - 5t, 0 \le t \le 1$

39 $x = \cos t, y = \sin t, z = 0, 0 \le t \le \pi$

41 Two arcs:
$\vec{r}\,(t) = 5\vec{i} + 5(-\cos t\vec{i} + \sin t\vec{j}\,)$, $0 \le t \le \pi$ or
$\vec{r}\,(t) = 5\vec{i} + 5(\cos t\vec{i} + \sin t\vec{j}\,)$, $\pi \le t \le 2\pi$

43 $\vec{r}\,(t) = (2 + 10t)\vec{i} + (5 + 4t)\vec{j}$

45 $\vec{r}\,(t) = (2 + ((t-20)/10)10)\vec{i}$
$+(5 + ((t-20)/10)4)\vec{j}$

47 $\vec{r}\,(t) = (2 - 10t)\vec{i} + (5 - 4t)\vec{j}$

49 No

51 (b) $-\vec{i} - 10\vec{j} - 7\vec{k}$

(c) $\vec{r} = (1-t)\vec{i} + (3 - 10t)\vec{j} - 7t\vec{k}$

53 (a) $\vec{r} = (\vec{i} + 3\vec{j} + 7\vec{k}\,) + t(2\vec{i} - 3\vec{j} - \vec{k}\,)$

(b) $(3, 0, 6)$

(c) $\sqrt{14}$

55 (a) $2\vec{i} - 5\vec{j} + 3\vec{k}$

(c) $x = 1 + 2t, y = -1 - 5t, z = 1 + 3t$

57 Different lines

59 Different lines

61 (a) $(-1, 4, -2)$

(b) $-\vec{i} - \vec{j} + 2\vec{k}$; other answers possible

(c) $\vec{r} = -\vec{i} + 4\vec{j} - 2\vec{k} + t(-\vec{i} - \vec{j} + 2\vec{k}\,)$

63 $x = -\frac{3}{2}t + \frac{5}{2}, y = \frac{1}{2}t + \frac{1}{2}, z = t.$

65 $x = -4, y = 2 + t, z = 3 + t$

67 (a) Straight lines

(b) No

(c) $(1, 2, 3)$

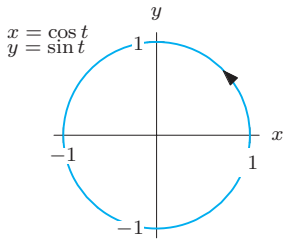
69 (a) Repeats every year

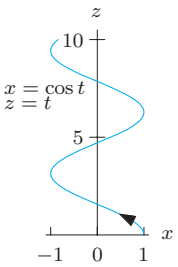
(b) Mid-August

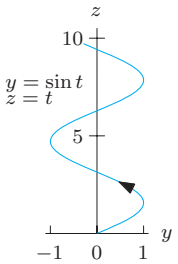
(c) Mid-April

(d) 2°C per month

71 Circle, cosine, sine







73 (a) $x = 2 + at, y = 1 + bt, z = 3 + ct$
(b) $a/2 = b/1 = c/3$

75 α is radius of cylinder
β determines vertical distance moved

77 (a) (vii)
(b) (ii)
(c) (iv)

79 $\vec{r} = 9.65\vec{k} + t(325\vec{i} + 563\vec{j} - 0.84\vec{k})$

81 (a) Parallel
(b) (i) Perpendicular
(ii) Parallel

83 Different parameterization of same curve

85 $x = \cos t, y = \sin t, z = 0$
$x = 0, y = \cos t, z = \sin t$

87 $x = t^3, y = 2t^3, z = 3 + 4t^3$

89 True

91 True

93 True

95 True

97 False

99 False

Section 17.2

1 $\vec{v} = 3\vec{i} + \vec{j} - \vec{k}, \quad \vec{a} = \vec{0}$

3 $\vec{v} = \vec{i} + 2t\vec{j} + 3t^2\vec{k}, \quad \vec{a} = 2\vec{j} + 6t\vec{k}$

5 $\vec{v} = -3\sin t\vec{i} + 4\cos t\vec{j},$
$\vec{a} = -3\cos t\vec{i} - 4\sin t\vec{j}$

7 $\vec{v} = \vec{i} + 2t\vec{j} + 3t^2\vec{k},$
Speed $= \sqrt{1 + 4t^2 + 9t^4},$
Particle never stops

9 $\vec{v} = 6t\vec{i} + 3t^2\vec{j},$
$\|\vec{v}\| = 3|t| \cdot \sqrt{4 + t^2},$
Stops when $t = 0$

11 $\vec{v} = 6t\cos(t^2)\vec{i} - 6t\sin(t^2)\vec{j},$
$\|\vec{v}\| = 6|t|,$
Stops when $t = 0$

13 Length $= \sqrt{42}$

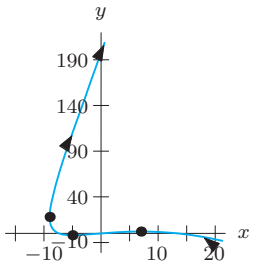
15 Length $= e - 1$

17 $\vec{v} = -6\pi\sin(2\pi t)\vec{i} + 6\pi\cos(2\pi t)\vec{j},$
$\vec{a} = -12\pi^2\cos(2\pi t)\vec{i} - 12\pi^2\sin(2\pi t)\vec{j},$
$\vec{v} \cdot \vec{a} = 0, \|\vec{v}\| = 6\pi, \|\vec{a}\| = 12\pi^2$

19 Line through $(2, 3, 5)$ in direction of
$\vec{i} - 2\vec{j} - \vec{k},$
$\vec{v} = 2t(\vec{i} - 2\vec{j} - \vec{k}), \vec{a} = 2(\vec{i} - 2\vec{j} - \vec{k})$

21 $x = 1 + 2(t - 2), y = 2, z = 4 + 12(t - 2)$

23 Vertical: $t = 3$
Horizontal: $t = \pm 1$
As $t \to \infty, x \to \infty, y \to \infty$
As $t \to -\infty, x \to \infty, y \to -\infty$



25 (a) $\vec{v}(2) \approx -4\vec{i} + 5\vec{j},$
Speed $\approx \sqrt{41}$
(b) About $t = 1.5$
(c) About $t = 3$

27 (a) $x = 2 + 0.6t, y = -1 + 0.8t, z = 5 - 1.2t, 0 \le t \le 5$
(b) $x = 2 + 1.92t, y = -1 + 2.56t, z = 5 - 3.84t, 0 \le t \le 1.56$

29 (a) 6.4 meters
(b) 1.14 sec
(c) 15.81 m/sec
(d) $(11.4, -5.7, 0)$
(e) -9.8 m/sec^2

31 (a) 5 secs; $(10, 15, 100)$
(b) $t = 0$, 10 secs, $\sqrt{113}$ cm/sec
(c) 5 secs, $\sqrt{13}$ cm/sec

33 (a) $t = 5.181$ sec
(b) $x = 103.616$ meters
(c) 2 meters
(d) 9.8 meters/sec^2
(e) $\theta = 0.896$; $v = 32.016$ meters/sec

35 (a) (IV); 4.5 sec; (0, 8.9 m, 0)
(b) (II); 3.2 sec; base of tower
(c) (V); 10 sec; halfway up

37 $\vec{r}(t) = 22.1t\vec{i} + 66.4t\vec{j}$
$+ (442.7t - 4.9t^2)\vec{k}$

41 (a) No
(b) $t = 5$
(c) $\vec{v}(5) \approx 0.959\vec{i} + 0.284\vec{j} + 2\vec{k}$
(d) $\vec{r} \approx 0.284\vec{i} - 0.959\vec{j} + 10\vec{k}$
$+ (t - 5)(0.959\vec{i} + 0.284\vec{j} + 2\vec{k})$.

43 (a) $\vec{r}(t) = t\cos(2\pi t)\vec{i} + t\sin(2\pi t)\vec{j},$
$0 \le t \le 100$
(b) $\vec{v} = \vec{r}\,'(t) =$
$(\cos(2\pi t) - 2\pi t\sin(2\pi t))\vec{i} +$
$(\sin(2\pi t) + 2\pi t\cos(2\pi t))\vec{j}$
$\|\vec{v}\| = (1 + 4\pi^2 t^2)^{1/2}$ cm/sec
(c) $\vec{a} = \vec{v}\,'(t) =$
$(-4\pi\sin(2\pi t) - 4\pi^2 t\cos(2\pi t))\vec{i} +$
$(4\pi\cos(2\pi t) - 4\pi^2 t\sin(2\pi t))\vec{j}$
$\|\vec{a}\| = 4\pi(1 + \pi^2 t^2)^{1/2}$ cm/sec^2

45 (a) $x \approx 694.444t$
$+ 20\cos(2\pi t)$cm
$y = 30 + 20\sin(2\pi t)$ cm
(c) At least 5.526 revs/sec

47 No

49 (a) $2\vec{r} \cdot d\vec{r}/dt$
(b) $\vec{a} \times d\vec{r}/dt$
(c) $r^3 d\vec{r}/dt + 3r^2\vec{r}$

55 Acceleration is a vector, not a scalar

57 $\vec{r}(t) = \left(t + 2t^2\right)\vec{i} + 2t\vec{j} + 3t^2\vec{k}$

59 False

61 False

63 False

65 False

67 True

69 False

Section 17.3

1 $\vec{V} = x\vec{i}$

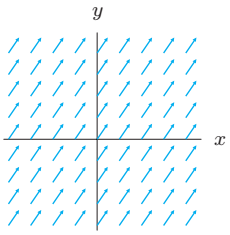
3 $\vec{V} = x\vec{i} + y\vec{j} = \vec{r}$

5 $\vec{V} = -x\vec{i} - y\vec{j} = -\vec{r}$

7 (a) y-axis
(b) Increasing
(c) Neither

9 (a) x-axis
(b) Increases
(c) Decreases

11



13

